

46-929, Fall 2019: Homework #6

Due 5:30 PM, Wednesday, December 11

1. This exercise is from Tsay (2010).

Consider the mean-zero AR(3) model

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \phi_3 X_{t-3} + a_t$$

where the a_t are iid $N(0, \sigma_a^2)$. Suppose that the observed data are

$$Y_t = X_t + \epsilon_t$$

where the ϵ_t are iid $N(0, \sigma_\epsilon^2)$.

(a) Write this model in state space form.

(b) The file `m-ppiaco4709.txt` (available on Canvas) contains year, month, day, and U.S. producer price index (PPI) from January 1947 to November 2009. The index is for all commodities, and not seasonally adjusted. Let $z_t = \log(Z_t) - \log(Z_{t-1})$, where Z_t is the observed monthly PPI. It turns out that an AR(3) model is adequate for z_t if the minor seasonal dependence is ignored. Let y_t be the sample mean-corrected series of z_t .

Suppose that y_t has independent measurement errors so that $y_t = x_t + \epsilon_t$ where x_t is a mean-zero AR(3) process and $V(\epsilon_t) = \sigma_\epsilon^2$. Use the state space model to estimate the parameters. Write down the fitted model. Also, create a plot showing the filtered estimates of x_t as a function of t .

2. In this exercise we will construct a time-varying coefficient model for a **pair** of series. The setup is much like that of the CAPM example from lecture.

Here, consider that we have two excess return series (for two different equities). We model the series as follows:

$$\begin{aligned} r_{1t} &= \alpha_{1t} + \beta_{1t} R_{M,t} + \epsilon_{1t} \\ r_{2t} &= \alpha_{2t} + \beta_{2t} R_{M,t} + \epsilon_{2t} \end{aligned}$$

where r_{it} is the excess return for equity i at time t and $R_{M,t}$ is the excess market return at time t .

If I combine the two ϵ into a single, two-dimensional vector, then we assume that this vector is bivariate normal with mean zero and is such that

$$V(\epsilon_{1t}) = V(\epsilon_{2t}) = \sigma_\epsilon^2$$

and that

$$\text{Cov}(\epsilon_{1t}, \epsilon_{2t}) = \gamma_\epsilon.$$

Next, we assume that for $i = 1, 2$,

$$\alpha_{i(t+1)} = \alpha_{it} + \eta_{it}$$

and

$$\beta_{i(t+1)} = \beta_{it} + \tau_{it}$$

where the pair (η_{1t}, η_{2t}) is bivariate normal with mean zero such that

$$V(\eta_{it}) = \sigma_\eta^2$$

and that

$$\text{Cov}(\eta_{1t}, \eta_{2t}) = \gamma_\eta.$$

Finally, we assume that the pair (τ_{1t}, τ_{2t}) is bivariate normal with mean zero and

$$V(\tau_{it}) = \sigma_\tau^2$$

and that

$$\text{Cov}(\tau_{1t}, \tau_{2t}) = \gamma_\tau.$$

So, the model has the following six parameters: σ_ϵ^2 , γ_ϵ , σ_η^2 , γ_η , σ_τ^2 , and γ_τ .

Place this model in state space form. In other words, specify the vectors $\mathbf{Y}_t$ and $\mathbf{X}_t$ and the matrices $\mathbf{F}_t$, $\mathbf{G}$, $\mathbf{W}$, $\mathbf{V}$.

Hint: In this case $\ell = 4$ and $k = 2$.